RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS8501 Theory of Computation

Practice's followed: Learning by Doing

Type of Activity: Active Learning

Topic: Turing Machine

Date: 17-09-19

Objectives:

O1: To make the students to design the Turing Machine for the given any language.

Justification for choosing the particular topic:

Turing Machine is an important problematic concept. Repetition in University exam question paper induces me to give the activity in class.

Activity Description:

Learning by Doing is an educational approach to problem-based learning.

Steps:

- Understand the problem
- Solve the problem by setting up an equation, drawing a diagram
- look back (check and interpret)

In the previous class problems on Turing Machine were solved. The same type of problems was given to the students. In the next class, some students were explained and solved the given problem in front of the class mates. Students were interactively asked the questions about solving steps and compared the answers. Finally answers were discussed.

Reflection critique
Post Implementation:

- This activity helped the student to analyze their performance of problem solving with others and it also induces the student to do problem daily.

Observation Made:

- Some of the students finding difficulties for constructing Turing Machine.
- In the construction phase of Turing machine(TM) for subtraction some of the students made mistakes in self-loop.

Steps taken for challenges:

- Additional problems were discussed.
- Students finding difficulties for constructing TM were identified and plan to give coaching in special Class.

Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	2	2		2					1			1

References:

https://uwaterloo.ca/centre-for-teaching-excellence/teaching-resources/teaching-tips/developing-assignments/cross-discipline-skills/teaching-problem-solving-skills
https://k12.thoughtfullearning.com/blogpost/teaching-innovation-and-problem-solving

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